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MR-Based PDE parameter estimation for Real-Time Adaptive Patient-Specific Chemotherapy for Brain Cancer

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ABSTRACT Brain cancer is a highly lethal disease, with malignant cases exhibiting a mortality rate of 64.3Accurately modeling tumor progression and the interactions between tumor cells, healthy tissue, and therapeutic drugs is essential for improving treatment outcomes.
To address this problem, this paper builds upon the model introduced by Ansarizadeh and presents a novel spatiotemporal partial differential equation that is solved using dynamic initial and boundary conditions estimated for each patient. To enable the estimation, the paper introduces the first ever dataset for predicting initial and boundary conditions for the proposed PDE model using a patient CT/MRI scan. Afterwards, the paper implements the proposed model to generate a patient-specific adaptive treatment drug plan for patients, where drug administration is dynamically applied only when a propagating tumor front is detected in specific spatial regions. To enable accuracy, images of the patient are taken periodically and fed into the pipeline to generate a more accurate and reliable treatment plan.

To efficiently solve this high-dimensional PDE system in a drug delivery embedded system, the paper employs Physics-Informed Neural Networks (PINNs) and DeepONet architectures, enabling fast and accurate simulations compared to traditional solvers such as FEM, FVM, or method of lines. The proposed framework thus integrates PDE modeling, real-time feedback control, and advanced neural PDE solvers, providing a more responsive, spatially targeted, and biologically realistic approach for simulating tumor progression and designing adaptive drug delivery strategies.

1 INTRODUCTION

1.1 Overview of Brain Tumors

THERE are more than 120 different types of brain tumors, lesions and cysts, which are differentiated by where they occur and what kinds of cells they are made of. Certain types of tumors are typically benign (noncancerous) like Meningioma, while others are typically malignant (cancerous) like Chordoma. Others may have a variable chance of being cancerous like Glioma [12]. Brain and CNS cancers pose a serious global health burden. In 2022, there were about 321,600 new cases and 248,400 deaths worldwide, with an age-standardized mortality rate of 2.6 per 100,000. The disease has a high mortality-to-incidence ratio—around 77.8in individuals aged 15 and older versus 47.7in children—highlighting the aggressive course of adult brain tumors and the difficulty in improving survival despite available treatments [12]. Historical data show a clear rise in the burden of brain and CNS cancers. Between 1990 and 2021, global age-standardized incidence rates increased from 3.75 to 4.28 per 100,000 (a 14.1rise), while prevalence grew from 8.66 to 12.01 per 100,000. In 2019, there were about 348,000 new cases and 246,000 deaths worldwide. The disease also shows notable sex differences: males accounted for 54of new cases and had higher mortality rates (3.54 per 100,000) compared to females (2.62 per 100,000) [8].

1.2 Imaging and Tumor Detection

Most of the brain is separated from the blood by the blood–brain barrier (BBB) that exerts a much more restrictive control over substances that are allowed to pass (or may even be subject to facilitate transport) than most other organs. Thus, many tracers that easily reach tumors in the body would reach brain tumors only once the tumor caused a disruption of the BBB. Such tumors that disrupt the BBB are usually detected in contrast-enhanced CT or MR scans. There are also brain tumors with an intact BBB, which are easily detected by increased signal on T2-weighted MRI scans without contrast enhancement [5]. Therefore, MR images are regarded as the standard for detection of brain tumors.

1.3 Current Treatment Approaches

1.3.1 Chemotherapy and Radiotherapy

While some works show that chemotherapy ineffective in high-grade gliomas [14], others show that chemotherapy produces significant results when combined with radiation. The authors in [3] showed that treatment with both chemotherapy and radiation produced significant results compared to treatment with radiation alone. They concluded that chemotherapy is advantageous for patients with malignant gliomas and should be considered part of the standard therapeutic regimen.

1.3.2 Immunotherapy

Another approach for brain tumor treatment is immunotherapy, where drugs focus on enhancing the immune system's efficiency in killing brain tumors [7]. Several studies have shown the effectiveness of immunotherapy with traditional therapies like chemotherapy. In such a study, Reardon et al demonstrated that bevacizumab and pembrolizumab were safely used together in patients with recurrent GBM [18].

1.4 Mathematical Modeling of Tumor Growth

Mathematical modeling offers a valuable approach to better understand tumor growth dynamics and optimize therapeutic treatments. Early mathematical descriptions often used ordinary differential equations (ODEs) to capture tumor cell population changes over time. While useful, these models didn't account for the spatial spread of tumor cells. To incorporate spatial heterogeneity and diffusion-driven invasion, partial differential equations (PDEs) have been increasingly employed. Reaction-diffusion models, in particular, have proven effective in describing the spatiotemporal evolution of tumor cell concentration and its interaction with surrounding healthy tissue [1].

1.4.1 Reaction–Diffusion Models

Noviantria et al. employed spatiotemporal reaction-diffusion equations incorporating exponential (linear), logistic, and Gompertzian (nonlinear) growth functions [13]. Using the finite difference method with the Crank-Nicolson scheme, these models captured distinct tumor growth patterns, revealing that Gompertzian growth reaches maximum concentration levels approximately 2.8 times faster than logistic or exponential models. Yuan et al. added a viscous stress tensor into the reaction-diffusion equation of the glioma growth model to account for the adhesion between glioma and normal cells [24]. The model employed a weighted parameter to control diffusion between gray and white matter. The simulation results demonstrated that the model produces anisotropic growth patterns that align with the glioma growth pattern. The model had better results than other models referenced in the paper.

1.4.2 Continuum–Mechanical and Porous Media Models

Other patient-specific continuum-mechanical models have leveraged the Theory of Porous Media (TPM) to simulate tumor growth and regression more realistically [22]. Multimodal MRI data inform these models by providing geometric representation (via 3D reconstructions) to define initial conditions related to tumor size and shape, material properties to parameterize tissue behavior in PDE formulations, and boundary conditions at tumor-healthy tissue interfaces. The PDEs are transformed into weak formulations and solved using FEniCS, enabling efficient finite element-based numerical simulations. This approach improves biological realism compared to population-based reaction-diffusion models. Tunç et al. conducted a comparison between three glioma growth models to focus on the mass effect: a Reaction-Diffusion-Advection model, which accounts for Mass effect (RDAM), a Reaction-Diffusion model with consistent Mass effect (RDM), and a Reaction-Diffusion model with no mass effect (RD) [23]. The models were calibrated with MRI data obtained during tumor development in mice. The paper showed that the RDM model most accurately predicts tumor growth, while the RDAM model presents the least variation in its estimates of the diffusion coefficient and proliferation rate.

1.5 Modeling Treatment Dynamics

Other studies have modeled the effect of chemotherapy on tumors. Rockne et al. developed a patient-specific continuum model that links MRI-derived tumor distributions to spatiotemporal tumor dynamics and treatment response [19]. Their framework augments diffusion and net proliferation with treatment terms to represent cytotoxic effects of therapy. Pérez-Aliacar et al. combined in vitro experiments with a mathematical model that explicitly tracks susceptible and drug-adapted subpopulations to investigate temozolomide (TMZ) resistance dynamics [15]. Other studies have expanded beyond modeling chemotherapy towards dynamic and personalized treatment. Sanz-Garcia et al tracked treatment progress in real time through circulating tumor DNA (ctDNA), small fragments of tumor DNA found in the blood [20]. They have shown that changes in ctDNA often appear weeks before traditional imaging shows tumor response, enabling earlier adjustments to therapy. However, variability in test results and how often blood samples are taken remain challenges for routine clinical use. Dodek et al. used artificial neural networks to estimate uncertain drug behavior in the body (pharmacokinetics) and calculate patient-specific optimal doses using minimal testing [2]. Their simulations suggest this method reduces variability in side effects compared with standard fixed-dose chemotherapy. Gallagher et al. derived optimal treatment timing strategies by matching tumor-growth models to patient-specific dynamics, showing that the frequency of clinical follow-up strongly influences adaptive control efficacy [4]. Building on this, Strobl et al. proposed the ADAPT framework (Adaptive Dosing Adjusted for Personalized Tumorscapes), integrating ctDNA, imaging, and PK/PD data into a dynamic decision-support loop [21]. Together, these studies mark a shift toward continuous, feedback-driven chemotherapy dosing capable of responding to tumor evolution in real time.

1.6 Image-Driven Parameter Estimation and Boundary Conditions

A different series of studies have attempted to estimate initial and boundary conditions as well as patient-specific parameters for solving well-known brain cancer models from medical images. Qiqi presented an image-driven reaction–diffusion modeling framework for simulating glioma tumor growth using multimodal MR images and histology-based initial conditions [16]. The model was solved under Neumann (no-flux) boundary conditions at the skull surface and with an initial condition derived from histological data. Liu et al. introduced a technique for estimating the volume of a brain tumor from MR images using fuzzy connectedness [9]. Their approach relied on implementing a customized standardization mechanism for each protocol of MR images The authors then segmented the tumor using fuzzy connectedness algorithms, where the user selects some initial points in the lesion and the algorithms run to complete the segmentation of the entire lesion and tumor. The final result of the paper is a lesion volume reporting in mm3, which serves as an important foundation for solving brain tumor models using reliable initial conditions for each patient. Hogea et al. introduced an approach for image-driven parameter estimation for PDE systems modeling brain tumors [6]. The authors estimated initial conditions through utilization of pre-segmented MR images of brain tumors and enforced Neumann boundary conditions. They estimated the PDE parameters through inverse PDE optimization. Given real-world data of tumor progression over a period of 2.5 years, the PDE equation that governs the diffusivity of a tumor is solved for coefficients that yield results matching those of real-world data. Martens et al. investigated whether routine magnetic-resonance imaging (MRI) data can provide reliable initial conditions through comparison against a non-operated glioblastoma brain [11]. They have concluded that estimating initial conditions from conventional MR images is limited in accuracy. However, their work only considered analysis of a single post-mortem case.

1.7 Proposed Study Framework

In this study, we build upon the reaction-diffusion model introduced by [1] and propose an enhanced spatiotemporal PDE framework that stimulates tumor progression and therapeutic drug interaction in patient-specific manner. The model incorporates dynamic initial and boundary conditions estimated from medical imaging data such as CT and MRI scans, allowing for personalized and biological consistent simulations. The proposed framework also integrates an adaptive drug delivery mechanism that dynamically adjusts treatment based on tumor progression patterns. Drug administration is applied selectively to spatial regions where the propagation tumor front is detected, minimizing unnecessary exposure to healthy tissues. Periodic medical imaging updates are incorporated to refine model parameters, improving the accuracy and responsiveness of treatment planning. To efficiently solve the high-dimensional PDE system, the study utilizes Physics-Informed Neural Networks (PINNs) and Deep Operator Networks (DeepONets) as fast and reliable alternatives to classical numerical solvers such as the Finite Element Method (FEM), Finite Volume Method (FVM), and Method of Lines (MOL). These neural architectures enable real-time approximation of complex PDE dynamics while maintaining physical consistency [17] [10].

2 The PDE Model

The PDE model was adopted from [1] and was modified to match the purposes of this project. The model was expanded from a single spatial dimension to 3 dimensions by changing the diffusion term to be a Laplacian in the 3 spatial coordinates. Therefore, the modified model is as follows:

$$\frac{\partial N}{\partial t} = r_2 N\left(1 - b_2 N\right) - c_4 T N - a_3\left(1 - e^{-U}\right) N + D_N \nabla^2 N, \quad (1)$$

$$\frac{\partial T}{\partial t} = r_1 T\left(1 - b_1 T\right) - c_2 I T - c_3 T N - a_2\left(1 - e^{-U}\right) T + D_T \nabla^2 T, \quad (2)$$

$$\frac{\partial I}{\partial t} = s + \frac{\rho I T}{\alpha + T} - c_1 I T - d_1 I - a_1\left(1 - e^{-U}\right) I + D_I \nabla^2 I, \quad (3)$$

$$\frac{\partial U}{\partial t} = v(t) - d_2 U + D_U \nabla^2 U. \quad (4)$$

The dependent variables $N, T, I,$ and U represent number of Normal, Tumor and Immune cells to the order of 10^{11} as well as the concentration of the chemotherapeutic drug. The diffusivity constants D_N, D_T, D_I, D_U were estimated from each patient based on their ADC image. The remaining constants were also adopted from [1] and their values are as follows:

TABLE 1. Model parameters and their biological interpretation

Parameter	Description	Value
a_1, a_2, a_3	Fraction cell kill	0.2, 0.3, 0.1
b_1, b_2	Carrying capacity	1, 0.81
c_1, c_2, c_3, c_4	Competition term	1, 0.55, 0.9, 1
d_1, d_2	Death rate	0.2, 1
r_1, r_2	Growth rate	1.1, 1
s	Immune source rate	0.33
α	Immune threshold rate	0.3
ρ	Immune response rate	0.2
$v(t)$	Time-dependent drug influx (weekly dosage)	Given weekly

The PDE was solved with Neumann zero flux boundary conditions at all boundaries. This can be justified because there is no movement nor transition of any cells at the edge of the human brain.

3 Methodology

3.1 Overall Framework

The proposed methodology integrates medical imaging, PDE-based tumor modeling, and neural PDE solvers within a closed-loop adaptive chemotherapy framework. Patient-specific MRI/CT images are first processed to estimate model parameters and spatial initial and boundary conditions. These estimates are then used to solve a coupled spatiotemporal PDE system describing tumor-immune-drug interactions. Periodic imaging updates provide feedback to adapt drug delivery in real time.

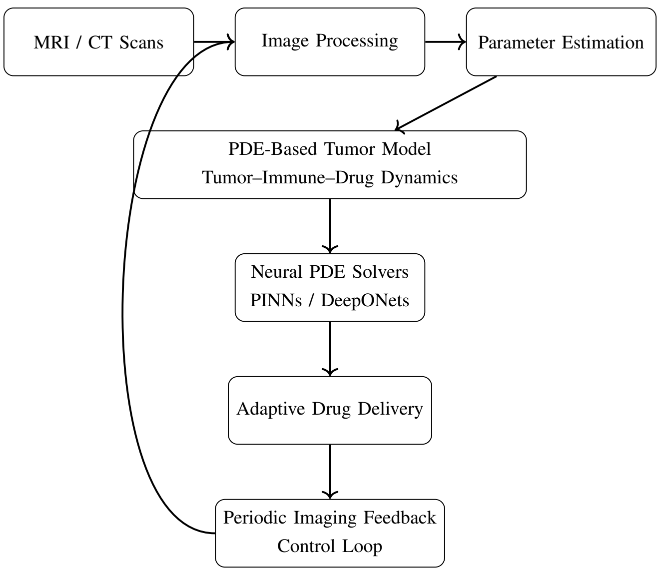

FIGURE 1. Overview of the proposed closed-loop adaptive chemotherapy framework.

3.2 Medical Image Processing and Parameter Estimation

3.2.1 Brain Segmentation

Brain tissue segmentation was performed using the IXI dataset, a publicly available collection of T1-weighted brain MRI scans acquired from healthy subjects. The dataset was divided into training, validation, and test sets. To reduce computational complexity while preserving anatomical context, center patches of size $80 \times 96 \times 112$ voxels were extracted from each 3D volume.

All input volumes were normalized using z-score normalization to standardize intensity distributions across subjects. The original multi-class anatomical labels were converted into a binary segmentation task, where all labels greater than zero were assigned to the brain tissue class, while zero represented non-brain regions such as background and skull.

A 3D U-Net architecture with three encoder–decoder levels was employed for brain segmentation. The encoder consisted of convolutional blocks with 32, 64, and 128 channels, each block including convolutional layers followed by batch normalization and ReLU activation functions to improve training stability. Spatial downsampling was achieved using max-pooling layers of size $2 \times 2 \times 2$ between encoder blocks. The decoder utilized transposed convolutions of size $2 \times 2 \times 2$ for upsampling, together with skip connections from the corresponding encoder layers to preserve spatial information. The final output layer applied a sigmoid activation function to generate a binary probability map representing brain tissue. The network contained approximately 4.4 million trainable parameters. The input tensor dimensions were $(1, 80, 96, 112)$, corresponding to single-channel volumetric MRI patches, and the output had the same spatial dimensions.

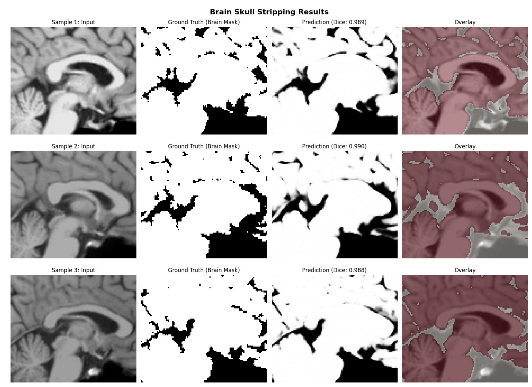

FIGURE 2. Example of brain tissue segmentation results. The predicted binary mask highlights the brain regions extracted from T1-weighted MRI patches.

To address class imbalance between brain and non-brain voxels, a combined loss function was used during training. The total loss was defined as a weighted sum of Dice Loss and Focal Loss:

$$\text{Loss} = 0.6 \times \text{Dice Loss} + 0.4 \times \text{Focal Loss} \tag{1}$$

Dice Loss was employed to measure the overlap between the predicted segmentation mask and the ground-truth mask. The Dice coefficient is defined as:

$$\text{Dice} = \frac{2 \cdot |P \cap T| + \epsilon}{|P| + |T| + \epsilon} \tag{2}$$

where P denotes the predicted mask, T denotes the ground truth, and $\epsilon = 1.0$ is a smoothing factor. Dice Loss is then computed as $1 - \text{Dice}$. Focal Loss was incorporated to reduce the influence of easy-to-classify voxels and focus training on more challenging regions. It is defined as:

$$\text{FL}(p_t) = -\alpha(1 - p_t)^{\gamma} \log(p_t) \tag{3}$$

where p_t is the predicted probability of the true class. The parameters were set to $\alpha = 0.25$ and $\gamma = 2.0$.

The model was trained using the AdamW optimizer with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-5}. A ReduceLROnPlateau learning rate scheduler was applied for 20 epochs. Training was performed with a batch size of 2 on a CUDA-enabled GPU, and the model with the highest validation Dice score was selected.

Segmentation performance was evaluated using the Dice Similarity Coefficient (DSC), Intersection over Union (IoU), precision, recall, specificity, and accuracy. Binary predictions were obtained by applying a threshold of 0.5 to the output probability maps.

3.2.2 Tumor Detection

Brain tumor detection and classification were performed using a curated brain MRI dataset, organized either as a 2-class problem (tumor versus no tumor). The dataset was divided into training, validation, and test sets to ensure robust evaluation and prevent overfitting. All images were resized to 224×224 pixels, converted to RGB format, and normalized using ImageNet statistics to maintain compatibility with the pre-trained network architecture.

We used transfer learning instead of training from scratch. We initialized ResNet-18 with ImageNet pre-trained weights, then replaced the last layer with a 2-class output and fine-tuned it on our MRI images.

The classification model was built using a ResNet-18 architecture, a deep residual convolutional neural network pre-trained on ImageNet. The pre-trained backbone was initially frozen to retain the learned feature representations, and the final fully connected layer was replaced with a custom linear layer mapping to the target number of classes. This fine-tuning strategy enabled the network to leverage pre-trained features while adapting to the specific tumor classification task.

Training was performed using the Adam optimizer with a learning rate of 1×10^{-4} and cross-entropy loss for multi-class classification. The model was trained for 15 epochs with a batch size of 16 images on a CUDA-enabled GPU. Both training and validation metrics were monitored at each epoch, and the model checkpoint achieving the highest validation accuracy was selected for evaluation.

Performance was evaluated on the held-out test set using accuracy, cross-entropy loss, and per-class performance metrics. Predictions were generated through the softmax output, allowing both the predicted label and associated confidence score to be reported for each image.

3.2.3 Tumor Segmentation

The dataset used is the BraTS 2020 training dataset, which provides multi-modal MRI scans of glioma patients with expert manual tumor annotations. For each subject, four MRI modalities were utilized: T1-weighted (T1), contrast-enhanced T1-weighted (T1ce), T2-weighted (T2), and FLAIR. The segmentation task focused on the whole tumor (WT) region. The original multi-class labels (0: background, 1: necrotic/non-enhancing core, 2: edema, 4: enhancing tumor) were converted into a binary segmentation problem, where all tumor-related labels (1, 2, 4) were merged into a single foreground class.

During preprocessing, the four MRI modalities were loaded and stacked as multi-channel inputs, and the ground-truth labels were binarized. To prevent data leakage, the dataset was split at the patient level (80:20 train-validation), ensuring that slices from the same patient did not appear in both sets. Each input sample had shape (channels, depth, height, width) = (4, 1, H, W), while the corresponding binary mask had shape (1, 1, H, W). The class imbalance problem was addressed by using binary segmentation, U-Net skip connections, and patient-level splitting to ensure exposure to diverse tumor appearances.

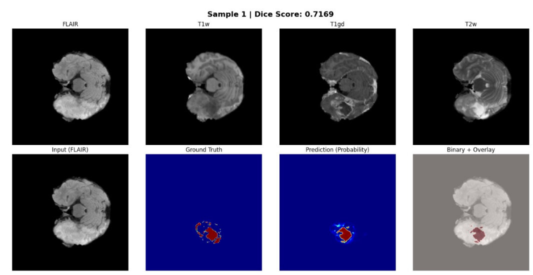

FIGURE 3. **Example of multi-modal MRI slices and the corresponding binary tumor segmentation mask.**

A multi-modal slice-wise 2.5D U-Net architecture was employed, using 3D convolutions with depth 1 to learn inter-modal relationships while operating on individual 2D slices. The encoder consists of three downsampling blocks with channels 32, 64, and 128, each containing two 3D convolution layers followed by GroupNorm and ReLU activation. The decoder uses trilinear upsampling with skip connections to recover spatial resolution. The network takes the four MRI modalities as input channels and outputs a single-channel binary segmentation mask through a $1 \times 1 \times 1$ convolution.

The model was trained using `BCEWithLogitsLoss` and the AdamW optimizer (weight decay = 1×10^{-5}) with a learning rate of 0.001. Training was conducted for 20 epochs with a batch size of 8. Mixed precision training using PyTorch's AMP was employed to reduce memory consumption and accelerate training, and the model was compiled with `torch.compile()` for additional efficiency.

3.2.4 MRI Modalities

Different MRI modalities provide complementary information about brain tissue and tumors:

- **T1-weighted (T1):** Provides high-resolution anatomical detail, with good contrast between grey and white matter.
- **FLAIR:** Highlights regions with edema or abnormal fluid accumulation by suppressing CSF signal, useful to delineate tumor extent.
- **Diffusion-Weighted Imaging (DWI):** Sensitive to water mobility, highlighting areas of restricted diffusion.
- **Apparent Diffusion Coefficient (ADC):** Quantitative map derived from DWI, representing voxel-wise water diffusivity. Lower ADC indicates higher cellular

density, higher ADC suggests necrotic or edematous regions.

TABLE 2. Comparison of MRI modalities for brain tumor analysis.

Modality	Main Use / Information Provided
T1	Anatomical detail, grey/white matter contrast
FLAIR	Highlights edema, abnormal fluid regions
DWI	Detects areas of restricted diffusion
ADC	Quantitative diffusion map, tumor cellularity

These modalities together allow accurate tumor segmentation and estimation of diffusion coefficients for patient-specific modeling.

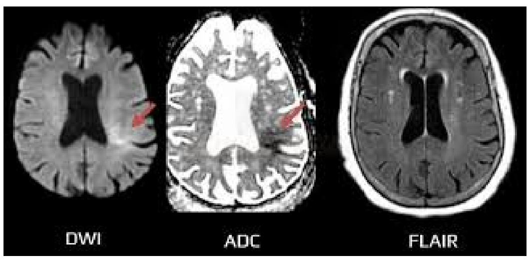

FIGURE 4. Different MRI modalities .

3.2.5 Diffusion Analysis

Apparent Diffusion Coefficient (ADC) maps obtained from diffusion-weighted MRI were used to estimate diffusion coefficients within brain tumors. ADC provides a voxel-wise measurement of water diffusivity, reflecting tissue cellularity and microstructure: lower ADC values typically indicate higher cellular density, whereas higher values suggest necrotic or edematous regions.

Tumor voxels were extracted using a binary segmentation mask, considering only voxels with mask value ¿ 0. Negative ADC values, likely due to noise, were excluded to ensure physiologically meaningful measurements. For each tumor, mean and median ADC values were calculated to represent overall diffusion characteristics, and the range of ADC values was reported to assess intra-tumoral heterogeneity.

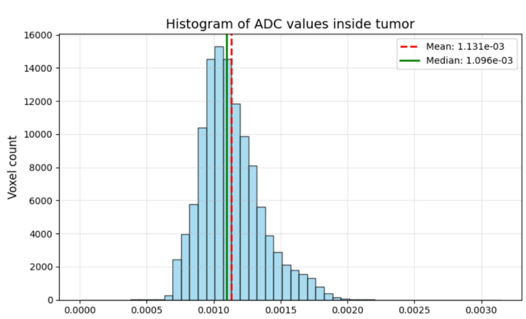

FIGURE 5. Histogram of tumor ADC values. The distribution highlights the mean and median diffusion coefficients, indicating intra-tumoral homogeneity with minor regions of increased diffusivity corresponding to necrotic or cystic areas.

The ADC values within the tumors ranged from 0.64 to 3.02×10^{-3} mm²/s, with a mean of 1.33×10^{-3} mm²/s and a median of 1.32×10^{-3} mm²/s. The ADC distribution was relatively narrow (range $\pm 0.2 \times 10^{-3}$ mm²/s), indicating a mostly homogeneous tumor structure. Regions of necrosis or cystic changes showed higher ADC values, slightly increasing the range.

TABLE 3. Typical ADC ranges for normal brain tissue and brain tumors.

Tissue / Tumor Type	ADC ($\times 10^{-3}$ mm²/s)
Normal White Matter	0.7 – 0.9
Normal Grey Matter	0.7 – 0.9
Low-grade Gliomas	1.2 – 1.5
High-grade Gliomas	0.8 – 1.2
Metastases	0.9 – 1.3

These measurements suggest that the tumor exhibits diffusion characteristics consistent with high-grade gliomas, with the mean ADC aligning with previously reported ranges for this tumor type. The small intra-tumoral ADC range indicates relative homogeneity, with only minor areas of increased diffusivity corresponding to necrotic or cystic regions.

3.3 Initial and Boundary Conditions

3.3.1 Initial Conditions

The metadata of the BraTS 2020 dataset indicates isotropic voxel spacing ($s_x = s_y = s_z = 1$ mm), implying that each 2D pixel implicitly represents a volumetric element of $1\ \text{mm}^3$. According to neurobiological and glioma modeling literature, approximately 10^5 brain cells occupy a $1\ \text{mm}^3$ volume. This assumption is used to convert segmentation outputs into biologically meaningful initial conditions.

The initial tumor cell density $T(x, 0)$ is obtained from the tumor volume predicted by the proposed 2.5D U-Net segmentation model. The segmented tumor volume is multiplied by a constant scaling factor of 10^5 cells per voxel to estimate the initial tumor cell distribution.

Similarly, the initial healthy brain cell density $N(x, 0)$ is derived from the brain segmentation produced by a 3D U-Net model. The segmented brain volume is scaled by the same factor of 10^5 cells per voxel to obtain the initial distribution of normal brain cells.

The initial immune cell population is assumed to be consistent with the parameter values reported in the reference tumor–immune interaction model used in this study.

The initial chemotherapeutic drug concentration $C(x, 0)$ is set to zero throughout the spatial domain, assuming that patients have not received chemotherapy at the initial time point.

3.3.2 Boundary Conditions

No-flux (Neumann) boundary conditions are imposed for all state variables at the brain boundary. This assumption reflects the confined anatomical structure of the brain and ensures that tumor cells, immune cells, and chemotherapeutic agents do not diffuse the brain domain.

3.4 The numerical scheme

The system was solved on a uniform $30 \times 30 \times 30$ Cartesian grid. Spatial derivatives were approximated using a second-order central difference scheme for the Laplacian operator:

$$\nabla^2 \phi_{i,j,k} \approx \frac{1}{\Delta x^2}\Big(\phi_{i+1,j,k} + \phi_{i-1,j,k} + \phi_{i,j+1,k} + \phi_{i,j-1,k} + \phi_{i,j,k+1} + \phi_{i,j,k-1} - 6\phi_{i,j,k}\Big). \tag{4}$$

Time integration was performed using the Method of Lines with a Forward Euler step of $\Delta t = 10^{-3}$. To ensure the accuracy of the solver, another independent PDE-solving library was utilized [25].

3.5 The PINN network

The input layer of the network consists of 4 inputs: t, x, y and z. The network has a depth of 6 layers and a width of 128 neurons per layer. The output layer consists of 4 neurons corresponding to the dependent variables. Softplus function is applied to the output of each neuron to ensure positive outputs. The Sigmoid Linear Unit (SiLU) was chosen as an activation function. The PINN is trained to satisfy the system of aforementioned PDEs. Let f_N, f_T, f_I and f_U denote the right-hand sides of the PDEs in equations 1-4. Let $N_\theta(t,x,y,z)$ be the neural network's prediction for N at time t and the coordinates x, y, z, where theta represents all trainable weights and biases. The same notation is assumed for T_θ ,I_θ and U_θ. Four types of losses were used to guide the network into the correct solution. The first loss is the physics residual, describing how well the model fits the structure of the equation. For a particular collection point t_c, automatic differentiation was used to compute the time derivative of the network's output at that point, e.g. $\frac{\partial N}{\partial t}|_{(t=t_c)}$. The N physics residual $R_N(t,x,y,z)$ is then formed as $R_N(t,x,y,z) = \frac{\partial N}{\partial t}|_{(t=t_c)} - f_N(t_c, x_c, y_c, z_c)$ Where $f_N(t_c)$ is the R.H.S of equation 1 estimated at $T(t_c, x_c, y_c, z_c)$, $I(t_c, x_c, y_c, z_c)$, and $U(t_c, x_c, y_c, z_c)$. Similar procedures were followed to estimate R_T,R_I and R_U. Since this residual is the difference between L.H.S and R.H.S at any of equations [1-4], the ideal value for it must be 0. To ensure the continuous accuracy of the model at any collection point, a set of 4096 collection points from $t = 0$ to $t = 14$ and in the interval $-2 \leq x, y, z \leq 2$ were used to estimate the physics loss in the loss function across the entire time region of interest.

The second loss is the data loss. For each of the known data points, the solution at that data point was compared to the model's prediction using mean squared error. The result of the numerical scheme of py-pde is assumed to approximate the true solution. This data loss helped in guiding the model towards the true solution since physics loss was insufficient for convergence in this PDE system. The initialized scheme for numerical data was a grid of 30x30x30 spatial points within the interval $-2 \leq x, y, z \leq 2$ and a timespan up to 14 with $dt = 0.001$.

The third loss is initial condition loss. A set of 100 points within the interval $-2 \leq x, y, z \leq 2$ were used to estimate the predicted initial condition by the model. The initial condition residual is then defined as the difference between that predicted initial condition and the provided initial grid for the model across the 100 points of estimation. Due to the volumetric nature of initial conditions in PDE, the PINN model was solved for only one initial condition that simulated physiological tumor initialization [1]. The initial conditions utilized in training were as following:

$$N(0,x,y,z) = 0.2\, \exp\big(-2x^2\big)\,, \qquad -2 \leq x \leq 2, \tag{5}$$

$$T(0,x,y,z) = 1 - 0.75\,(x), \qquad -2 \leq x \leq 2, \tag{6}$$

$$I(0,x,y,z) = 0.375 - 0.235\,^2(x), \qquad -2 \leq x \leq 2, \tag{7}$$

$$U(0,x,y,z) = (x), \qquad -2 \leq x \leq 2. \tag{8}$$

The fourth loss is the boundary condition loss. The model enforces Neumann zero flux boundary conditions at 200 points within the interval $0 < t \leq 14$. For a spatial coordinate k, the boundary condition for N is given by $\frac{\partial N}{\partial k}|_{k=boundary} = 0$. Therefore, the boundary residual of N is the difference between the estimated value through auto differentiation and zero.

The total loss of each of those four compartments was added through empirical weights that optimized training accuracy. The total loss is therefore given by:

$$L = \lambda_{\text{physics}} \cdot L_{\text{physics}} + \lambda_{\text{data}} \cdot L_{\text{data}} + \lambda_{\text{IC}} \cdot L_{\text{IC}} + \lambda_{\text{BC}} \cdot L_{\text{BC}} \tag{9}$$

Where $\lambda_{\text{physics}} = 10$, $\lambda_{\text{data}} = 100$, $\lambda_{\text{IC}} = 10$ and $\lambda_{\text{BC}} = 50$. The model training was performed on a standard workstation and took advantage of GPU acceleration for faster computation. The network was trained using AdamW Optimizer with an initial learning rate of 5×10^{-4} and a weight decay of 1×10^{-5}.

The model's performance was evaluated against the numerical data used in the training and the average MSE of each compartment was estimated.

3.6 The DeepONet network

The Deep Operator Network (DeepONet) architecture aims to approximate an operator by learining the realtion between the input and output functions of the system in a supervised approach.The operator:

$$\mathcal{G} : u_0(x,y,z) \mapsto u(t,x,y,z) \tag{10}$$

where $u_0(x,y,z)$ denotes the initial conditions fields at $t = 0$, and $u(t,x,y,z)$ denotes the solution of the PDE at an arbitrary point in space and time.

That operator will be used to predict the output function at a given input function. The input function of this model is the fields of initial conditions and parameters of the PDE system. The output function is the fields of dependent variables.

DeepONets is generally composed of two subnetworks: a branch network, which encodes the initial conditions magnitudes and system parameters into a feature representation, and a trunk network, which learns basis functions over the output domain. These two components are combined—typically via a dot product—to reconstruct the output function. The combination is governed by the universal approximation theory for nonlinear operators:

$$\mathcal{G}[u(t,x,y,z)] \approx \sum_{i=1}^{k} b_i(u) \cdot t_i(u) \tag{11}$$

where $b_i(u)$ is the branch network output, and $t_i(u)$ is the trunk network output.

This DeepONet was modified to handle a 3D grid of initial conditions via a CNN network to flatten the grid. The CNN applied a sequence of stride-2 convolutions with increasing channel depth. Each layer was followed by batch normalization and ReLU activation function. Kaiming initialization was used for convolutional weights while Xavier initialization was used for the linear layers. It returns 128 features for Tne branch network.

The branch network received the output of the CNN to process it through a MLP with four hidden layers of 256 neuron. Each layer was followed by layer normalization and a Tanh activation function.

The trunk processed the normalized spatiotemporal query points, with its input layer consisting of 4 neurons representing the space-time independent varaibles. It shares the same architectural structure as the branch network, including the number of layers, neurons per layer, and activation functions.

The training data simulated different cases of tumor cells diffusion patterns. Regions with high density of tumor cells were modeled as spherical and elliptic masses. It had smooth borders to like benign tumors while adding heterogeneity to include malignant tumors. that heterogeneity included: elongated shapes, multifocal shapes, irregular patterns, and introducing gaussian noise. Normal cells density was highly affected by tumor cells. Immune cells and drug were more concentrated at tumors borders. Different tumor stages were included by changing the radii of tumor regions and modifying their concentration accordingly.

The loss function is a mean squared error function:

$$\text{MSE}(y,\hat{y}) = \frac{1}{N}\sum_{i=0}^{N-1}(y_i - \hat{y}_i)^2$$

The AdamW optimizer was used with a learning rate of $5 \cdot 10^{-4}$ and a weight decay of $1 \cdot 10^{-5}$. A cosine annealing learning rate scheduler was used to prevent oscillations around suboptimal solutions.

3.7 Real-Time Adaptive Chemotherapy Strategy

The proposed framework incorporates a real-time adaptive chemotherapy strategy driven by imaging-derived diffusion information and PDE-based tumor dynamics. Apparent Diffusion Coefficient (ADC) maps obtained from diffusion-weighted MRI are used to estimate spatially varying diffusion coefficients $D(x)$, which reflect tumor cellularity and invasive potential. These coefficients are directly embedded into the tumor growth model, enabling patient-specific and spatially heterogeneous tumor behavior. At predefined time intervals, updated MRI and ADC measurements are incorporated to recalibrate the tumor state and refine the diffusion parameters. Based on predicted tumor growth, spatial spread, and local diffusion characteristics, chemotherapy dosage and spatial delivery are adaptively adjusted. Regions exhibiting lower ADC values, corresponding to higher cellular density and aggressive tumor behavior, receive increased therapeutic emphasis, while regions with higher ADC values are treated more conservatively to limit toxicity to healthy tissue.

This closed-loop feedback strategy enables dynamic adaptation of treatment to patient-specific tumor evolution. By integrating ADC-based diffusion modeling with real-time PDE predictions, the proposed framework supports personalized and optimized chemotherapy planning.

4 Results and Discussion

Table 4 shows the average MSE per compartment of the PINN network. The PINN Network demonstrated sufficient accuracy to the numerical solution. Figure 6 shows a sample comparison between the predictions of the PINN model and the numerical solution. Although the performance of the PINN was more than acceptable in terms of numerical error, the PINN model could not be generalized to solve for different initial conditions without retraining. This is mainly because the PINN networks accepts scalar values as inputs. An initial condition would mean passing a 3D matrix as an input to the network, which is not allowed within the framework of PINNs. This issue is solved in the DeepONet model, which can be attributed to the vector nature of its operation.

Compartment	MSE
N	0.5581
T	2.924×10^{-4}
I	3.346×10^{-5}
U	2.715×10^{-5}

TABLE 4. Mean Squared Error (MSE) for each PINN model compartment compared to ground truth.

Table 5 shows the mean square error of the DeepONet model predictions on the test cases. The model surpasses the PINN's accuracy in the N compartment while it provides acceptable accuracy in the remaining compartments. The key milestone is the ability of DeepONet to generalize the solution of the PDE model across different initial conditions and predicting unseen data with reliable accuracy. A sample

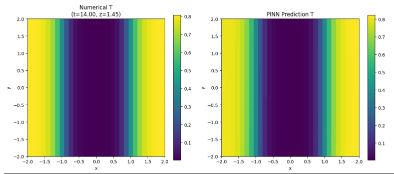

FIGURE 6. An XY slice of the numerical solution for the T compartment compared with the PINN Solution at z=1 and t=14

DeepONet solution of the unseen case compared to the numerical solution is provided in figure 7.

Compartment	MSE
N	7.675×10^{-4}
T	8.717×10^{-5}
I	1.199×10^{-2}
U	2.930×10^{-4}

TABLE 5. Mean Squared Error (MSE) for each DeepONet model compartment compared to ground truth.

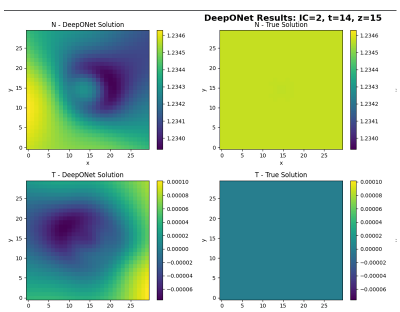

FIGURE 7. An XY slice of the numerical solution for the N and T compartments compared with the DeepONet Solution at z=15 and t=14

In terms of speed, the deep learning models showed significantly faster speeds at obtaining the grid solution after 14 weeks compared to traditional numerical solvers. This is demonstrated by the results in table 6. The PINN model was 2290 times faster than the fastest numerical solver. The DeepONet model was 59 times faster than the fastest numerical solver. This significance of speed in solving can be attributed to the vectorization feature of neural networks, which allows them to process batches of data at the same iteration.

Solution model	Solution time (s)
py-pde	93.1
method of lines	50.6
PINN	0.0221
DeepONet	0.853

TABLE 6. Time taken by each solution model to solve a 3D initial conditions grid

5 Conclusion

This work presented a patient-specific, image-driven cancer therapeutic planning approach that is governed by a coupled spatiotemporal PDEs. The work started by extending the adopted PDE to three dimensions to ensure applicability. Afterwards, the work introduced an approach for estimating initial and boundary conditions for each patient through segmentation of the brain and tumor within their MRI images as well as an approach to estimate patient-specific diffusivity from ADC maps from MRI images.

To provide the ability for repetitive analysis of the PDE at different conditions, the paper introduced two deep-learning based approaches for solving the PDE efficiently: A Physics Informed Neural Network and a Deep Operator Network. Both networks succeeded in adapting to the true solution with acceptable error, but the DeepONet was capable of generalizing beyond the training initial conditions. This allowed the DeepONet model to solve the PDE at unseen initial conditions with reasonable error, proving the model capable of fast analysis of the PDE without the need for retraining. The true solution was verified through method of lines as well as external independent solvers.

Overall, this paper proposes the workflow of analyzing the patient's initial condition and diffusivity coefficient's through the imaging and segmentation pipeline. Those initial conditions can then be analyzed numerically to obtain an estimation for the severity of the patient's case in 2 weeks. The deep learning models can then be utilized for rapidly solving thousands of initial cases involving treatment to find the case with minimal cancer diffusion in the future. This process can be repeated then regularly with new images, enabling an adaptive and patient-specific method for chemotherapy treatment of brain caner.

6 Code availability

The code for implementing this paper is available on the github link

7 Future Work

In subsequent iterations of this research, we expect to increase the number of initial conditions that the DeepONet model is trained upon to further increase the accuracy. We also aim to find an alternative way to calculate diffusivity from a single MRI image instead of relying on 3 different scanning techniques. We also want to generalize the DeepONet to produce solutions at different diffusivity values. Finally, we aim to add a physical prototype for implementing the proposed method in suggesting adaptive treatment plans.

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